Ascend: Work Breakdown Structure

Project	Ascend: AI-powered growth intelligence for small businesses
Document	Work Breakdown Structure
Course	P6 Special Topics
Team	Aaryaman Chopra, Ethan de Guzman, Palash Desarda, Kaiyaan Faisal
Repository	github.com/EDGDevelopment/Ascend (documents are in docs/deliverables)
Version	Draft 1.0, November 6, 2026

1. Overview

We split the project into six areas with 27 work packages in total. Areas 2 to 5 follow the roles we set at Sprint Review 1: Aaryaman on the model, Kaiyaan on data, Palash on the frontend and Ethan on the backend. Area 1 covers the course deliverables and area 6 covers work that involves more than one person. As of today, 13 packages are done, 7 are under way and 7 have not started. The package numbers are also used in the Project Schedule and the Functional Specification.

2. Work Packages

Each package has one owner, a status as of November 6 and the month it is planned for. Rows with a gray background are the six areas.

ID	Work package	Owner	Status	When
1	**Project management and deliverables**	**Everyone**		
1.1	Pitch deck and script	Everyone	Done	Before Oct 6
1.2	Sprint reviews	Everyone	Reviews 1 (Oct 6) and 2 (Oct 27) done; more to come	Oct to Jan
1.3	Functional spec, WBS and schedule (drafts)	Everyone	In progress	Due Nov 6
1.4	Google course certificates in docs/ (Building and Deployment, one pair per member)	Everyone	Done (8 PDFs in docs/)	Nov 6
1.5	Final demo and presentation	Everyone	Not started	Late Jan
2	**Financial data**	**Kaiyaan**		
2.1	Public financial datasets for training and testing	Kaiyaan	Done	Oct
2.2	Synthetic test files (14 CSVs, 3 of them meant to be rejected)	Kaiyaan	Done	Oct
2.3	Collect real small-business data	Kaiyaan	Not started	Nov
2.4	Clean and validate the real data	Kaiyaan	Not started	Nov
3	**Model**	**Aaryaman**		

ID	Work package	Owner	Status	When
3.1	Statistical forecasting engine (seasonal smoothing, last year scaled, Theta)	Aaryaman	Done	Oct
3.2	Patch Forecaster: train, export weights, TypeScript inference	Aaryaman	Done on public data	Oct
3.3	Monte Carlo ranges and accuracy self-test	Aaryaman	Done	Oct
3.4	Valuation, cash safety, break-even, what-if and alert maths	Aaryaman	Done; needs a second reader	Oct
3.5	Backtest, then score and tune on real data; fix ranges for short history	Aaryaman	Backtest done; tuning not started	Nov to Dec
4	**Frontend**	**Palash**		
4.1	Landing page and sign-in screens	Palash	Done	Oct
4.2	Dashboard shell and pages	Palash	Done	Oct
4.3	Financial model page: three-step flow and five tabs	Palash	Done	Oct
4.4	Excel and CSV import and export	Palash	Done	Oct
4.5	Polish: non-profit flow, UX fixes, second read of the maths	Palash	Not started	Nov
5	**Backend**	**Ethan**		
5.1	Supabase auth and profiles and businesses schema	Ethan	Done	Oct
5.2	Edge Function for the model	Ethan	Written, not deployed	Nov
5.3	Deploy and test the backend end to end	Ethan	Not started	Nov
5.4	Dataset storage (migration 0003) and upload pipeline	Ethan	Written, not deployed	Nov to early Jan
6	**Integration and quality**	**Shared**		
6.1	Dashboards running on real model output	Ethan, Palash	Mostly done	Early Jan
6.2	Automated tests (8 pass today; add valuation, cash and scenario tests)	Aaryaman	In progress	Nov to Jan
6.3	Security and privacy review: who can see client data and how it is stored	Everyone	Not started	Late Jan
6.4	README and docs refresh	Everyone	Not started	Nov

3. Ownership Summary

Nobody owns more than 6 of the 27 packages. Real data and backend deployment are the two things other work waits on.

Owner	Area	Packages	Done	Under way	Not started
Aaryaman Chopra	Model (3.1 to 3.5) and automated tests (6.2)	6	4	2	0

Owner	Area	Packages	Done	Under way	Not started
Kaiyaan Faisal	Financial data (2.1 to 2.4)	4	2	0	2
Palash Desarda	Frontend (4.1 to 4.5)	5	4	0	1
Ethan de Guzman	Backend (5.1 to 5.4)	4	1	2	1
Ethan and Palash	Dashboards on real model output (6.1)	1	0	1	0
Everyone	Pitch, reviews, deliverables, security review, docs (1.1 to 1.5, 6.3, 6.4)	7	2	2	3
Total		**27**	**13**	**7**	**7**

4. What Is Still Open

The build is ahead of plan and the real data is behind it. This matches what we said at Sprint Review 2: ahead on the build, behind on real small-business data, which is the focus for November.

1. Real data (2.3 and 2.4, Kaiyaan, November). The model cannot be tuned on real clients until the data arrives and is clean.
2. Backend deployment (5.3, Ethan, November). The Edge Function and migration 0003 are written but have never run on a live Supabase project.
3. Second read of the maths (4.5, Palash, November). About 5,000 lines of forecasting and valuation code have one author.
4. Tuning on real data (3.5, Aaryaman, December). This starts once 2.4 is done and includes fixing ranges for short histories.
5. Security and privacy review (6.3, everyone, late January). It needs 5.3 finished first so there is something live to review.
6. Drafts due today (1.3). This document, the Functional Specification and the Project Schedule go in the docs folder of the GitHub repo.